

Unit 2: First Order Linear Differential Equations — Practice Problems and Solutions

Prepared by Staples Education

Review Guide of Graded Problems with Complete Step-by-Step Solutions

Part II — Review Guide: Practice Problems

Work the following ten problems in order; they are graded from direct separation through linear factors, initial value problems, the homogeneous substitution, and a full physical model. Solutions follow in Part III.

1. Solve $\frac{dy}{dx} = \frac{x}{y}$ by separation of variables.
 2. Solve $\frac{dy}{dx} = e^{x-y}$.
 3. Solve $\frac{dy}{dx} = y \cos x$, and identify any solution that is lost when you divide by y .
 4. Solve the linear equation $\frac{dy}{dx} + 3y = 6$ using an integrating factor.
 5. Solve $x\frac{dy}{dx} + y = x^2$ by first writing it in standard linear form.
 6. Solve $\frac{dy}{dx} + \frac{2}{x}y = x$.
 7. Solve the initial value problem $\frac{dy}{dx} + 2y = 4$ with $y(0) = 1$.
 8. Solve the homogeneous equation $\frac{dy}{dx} = \frac{x+y}{x}$ using the substitution $y = vx$.
 9. Solve the homogeneous equation $\frac{dy}{dx} = \frac{x^2 + y^2}{xy}$ using $y = vx$.
 10. **(Application)** An object at 100° is placed in a room held at 20° . After 10 minutes its temperature is 60° . Find the temperature function $T(t)$ and determine how long it takes the object to reach 30° .
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Part III — Complete Solutions

Solution 1. Separate, then integrate.

$$\begin{aligned}y \, dy &= x \, dx \\ \frac{y^2}{2} &= \frac{x^2}{2} + C_1 \\ y^2 &= x^2 + C \quad (\text{general solution}).\end{aligned}$$

Solution 2. Split the exponential as $e^{x-y} = e^x e^{-y}$, then separate.

$$\begin{aligned}e^y \, dy &= e^x \, dx \\ e^y &= e^x + C \\ y &= \ln(e^x + C).\end{aligned}$$

Solution 3. Separate (assuming $y \neq 0$), integrate, and exponentiate.

$$\begin{aligned}\frac{1}{y} \, dy &= \cos x \, dx \\ \ln |y| &= \sin x + C \\ y &= C e^{\sin x}.\end{aligned}$$

Dividing by y assumed $y \neq 0$, so it dropped the equilibrium solution $y = 0$ — which is recovered as the special case $C = 0$.

Solution 4. Here $P = 3$, so the integrating factor is $\mu = e^{\int 3 \, dx} = e^{3x}$. Multiply through:

$$\begin{aligned}\frac{d}{dx}[e^{3x}y] &= 6e^{3x} \\ e^{3x}y &= \int 6e^{3x} \, dx = 2e^{3x} + C \\ y &= 2 + Ce^{-3x}.\end{aligned}$$

Solution 5. Divide by x to reach standard form:

$$\frac{dy}{dx} + \frac{1}{x}y = x, \quad \mu = e^{\int (1/x) \, dx} = e^{\ln |x|} = x.$$

Multiply through and integrate:

$$\begin{aligned}\frac{d}{dx}[xy] &= x \cdot x = x^2 \\ xy &= \frac{x^3}{3} + C \\ y &= \frac{x^2}{3} + \frac{C}{x}.\end{aligned}$$

Solution 6. With $P = \frac{2}{x}$, integrate the coefficient and exponentiate:

$$\mu = e^{\int (2/x) dx} = e^{2 \ln |x|} = x^2.$$

Multiply through and integrate:

$$\frac{d}{dx} [x^2 y] = x^2 \cdot x = x^3$$

$$x^2 y = \frac{x^4}{4} + C$$

$$y = \frac{x^2}{4} + \frac{C}{x^2}.$$

Solution 7. The integrating factor is $\mu = e^{\int 2 dx} = e^{2x}$. Multiply through and integrate:

$$\frac{d}{dx} [e^{2x} y] = 4 e^{2x}$$

$$e^{2x} y = 2 e^{2x} + C$$

$$y = 2 + C e^{-2x}.$$

Apply the initial condition $y(0) = 1$:

$$1 = 2 + C \implies C = -1$$

$$y = 2 - e^{-2x}.$$

Solution 8. Rewrite as $\frac{dy}{dx} = 1 + \frac{y}{x}$ and substitute $y = vx$, so $\frac{dy}{dx} = v + x \frac{dv}{dx}$:

$$v + x \frac{dv}{dx} = 1 + v$$

$$x \frac{dv}{dx} = 1 \implies dv = \frac{1}{x} dx$$

$$v = \ln |x| + C.$$

Back-substitute $v = \frac{y}{x}$:

$$y = x \ln |x| + Cx.$$

Solution 9. Substitute $y = vx$. The right side becomes

$$\frac{x^2 + v^2 x^2}{x \cdot vx} = \frac{1 + v^2}{v}.$$

With $\frac{dy}{dx} = v + x \frac{dv}{dx}$,

$$v + x \frac{dv}{dx} = \frac{1 + v^2}{v} = \frac{1}{v} + v$$

$$x \frac{dv}{dx} = \frac{1}{v} \implies v dv = \frac{1}{x} dx$$

$$\frac{v^2}{2} = \ln |x| + C_1 \implies v^2 = 2 \ln |x| + C.$$

Back-substitute $v = \frac{y}{x}$:

$$\frac{y^2}{x^2} = 2 \ln |x| + C \implies y^2 = x^2 (2 \ln |x| + C).$$

Solution 10. Newton's law of cooling with surrounding temperature $T_s = 20$ and initial temperature $T_0 = 100$ gives

$$T(t) = 20 + (100 - 20) e^{-kt} = 20 + 80 e^{-kt}.$$

Use the reading $T(10) = 60$ to find k :

$$60 = 20 + 80 e^{-10k} \implies 80 e^{-10k} = 40$$

$$e^{-10k} = \frac{1}{2} \implies -10k = -\ln 2 \implies k = \frac{\ln 2}{10}.$$

Hence the temperature function is

$$T(t) = 20 + 80 e^{-(\ln 2/10)t} = 20 + 80 \left(\frac{1}{2}\right)^{t/10}.$$

To reach 30° , solve $T(t) = 30$:

$$30 = 20 + 80 e^{-kt} \implies 80 e^{-kt} = 10 \implies e^{-kt} = \frac{1}{8}$$

$$-kt = -\ln 8 = -3 \ln 2 \implies t = \frac{3 \ln 2}{k} = \frac{3 \ln 2}{\ln 2/10} = 30 \text{ minutes}.$$

Unit 2 Key Takeaway

Every method in this unit reduces to a single move — rewrite the equation into a form you can integrate. **Separation of variables** splits a factored rate law; the **integrating factor** $\mu = e^{\int P dx}$ collapses a linear left side into one derivative; and the **substitution** $y = vx$ turns a homogeneous equation separable. Read the structure first, and the technique always follows.